Abu Dhabi City Campus Daily/Weekly Planner

Week 8: 19th – 23rd October 2026

Subject: Mathematics Diploma Programme **Unit Title: Number and Algebra** **Grade: 11**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: Parallel Lines	I can determine and compare the equations of parallel lines and use slope to justify why two lines are parallel.	2.1 Clarify slope, gradient and line-equation vocabulary using worked examples and visual graphs. 2.3 Support decoding of algebraic notation through multiple representations: equation, gradient-intercept form, graph and coordinate points.	Lesson notes, review sheet and resources can be found on Teams and OneNote. Linear functions [IB Maths AI SL/HL] Linear functions [IB Maths AI SL/HL]
Lesson 2: Perpendicular Lines	I can determine the equation of a perpendicular line and justify perpendicularity using gradients and the negative reciprocal relationship.	2.3 Support decoding of negative reciprocals and point-slope form using step-by-step worked examples. 5.3 Build fluency from identifying perpendicular slopes to constructing and verifying equations.	
Lesson 3: HL Introduction to Matrices – Addition & Subtraction	I can identify matrix dimensions and add or subtract matrices when their dimensions are compatible.	5.3 Build fluency through graduated practice: identify dimensions → determine compatibility → perform operations → interpret results. 8.2 Vary challenge through differentiated HL matrix-operation problems and error analysis.	
Lesson 4: Summative Review	I can apply and connect the skills from this unit to solve unfamiliar problems accurately and explain the mathematical reasoning used.	2.3 Support decoding of multi-step summative questions using worked examples and question-analysis routines. 5.3 Build fluency through targeted retrieval and graduated exam-style practice.	
Homework	Revision Village / InThinking targeted practice and teacher-provided review resources on Teams/OneNote.		
Teacher's notes/ Reminders	Students must bring a device, graphing calculator/GDC and notebook.		

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Technology Integration	Microsoft Teams; Desmos for graphing parallel/perpendicular lines; GDC for checking equations and matrices
Cross-curricular links	**Physics/Engineering:** perpendicular and parallel structures, coordinate modelling and vectors. **Computer Science/Data:** matrices as structured arrays and data representation.

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External Assessment Integration	Direct alignment with IB Mathematics AI
Wellbeing/ ATL Skills	Thinking Skills: analyse slope relationships, select appropriate representations and verify solutions. Communication Skills: use precise geometric and matrix terminology and justify conclusions. Self-Management Skills: organise multi-step work, manage time during review and reflect on errors. Social Skills: collaborate during problem-solving and provide constructive peer feedback.
Standards	IB Mathematics: Applications and Interpretation / HL. Geometry and Trigonometry: equations of lines, gradient, parallel and perpendicular relationships. HL: matrices — notation, dimensions, addition and subtraction. Mathematical communication: accurate notation, reasoning, verification and interpretation.
Inquiry/ TOK Question	How can mathematical representations such as equations, graphs and matrices help us describe relationships accurately and make reliable decisions?
Success Criteria (Measurable)	**Lesson 1 – Parallel Lines** • **All students must** identify gradients and determine whether two lines are parallel. • **Most students should** find the equation of a line parallel to a given line through a point and justify the result. • **Some students could** solve unfamiliar parameter problems involving parallel lines and justify their reasoning. **Lesson 2 – Perpendicular Lines** • **All students must** identify perpendicular gradients using negative reciprocals. • **Most students should** determine and verify the equation of a perpendicular line through a given point. • **Some students could** solve unfamiliar coordinate/parameter problems and justify perpendicularity algebraically and graphically. **Lesson 3 – HL Matrices** • **All students must** identify matrix dimensions and perform compatible addition/subtraction accurately. • **Most students should** determine whether matrix operations are possible and explain why. • **Some students could** analyse errors or solve unfamiliar HL matrix-operation problems with precise notation. **Lesson 4 – Summative Review** • **All students must** accurately retrieve and apply the key skills from the unit. • **Most students should** solve mixed exam-style problems and correct errors independently. • **Some students could** solve unfamiliar HL-style problems, compare methods and justify their choices.
	Area: Citizenship
	Implementation: Connect mathematical modelling and accurate spatial reasoning to UAE engineering, architecture, transport and infrastructure, emphasising precision and responsible decision-making.
	Implementation: Highlight how coordinate geometry and matrix representations support modern design, technology and data-driven systems.